Project: Porewater SO4/Cl

Month Year Samples

2026-10-16

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##Add Required Packages

0.1 Run Information - EDIT EACH RUN

```
###### Run information - PLEASE CHANGE
  Date_Run = as.Date("2026-07-30")  #Date that instrument was run
  Run_by = "Zoe Read"  #Instrument user
  Script_run_by = "Zoe Read" #Code user
  run_notes = "" #any notes from the run

  Project_name <- "COMPASS:Synoptic" #LTREB/SMARTX/GENX/COMPASS:TEMPEST
  Site <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names (TMP, F
  Site_pattern <- paste(Site, collapse = "|")

  Plot <- c("UP", "TR", "SWAMP", "WC", "SW") ##Use "ALL" for LTREB sample plots
  Plot_pattern <- paste(Plot, collapse = "|")

  #Set time zone
  common_tz = "Etc/GMT+5"
  Sys.setenv(TZ = "America/New_York")


##Fix samples that were entered incorrectly - PLEASE CHANGE
  #Enter samples to remove
  samples_to_remove <- NA

  #Enter IDs to change
  sample_ID <- c(NA)
  New_IDs <- c(NA)
  IDs_to_change <- data.frame(sample_ID, New_IDs)

##### Project-specific R Scripts ####
viz_script <- switch(
  Project_name,
  "COMPASS:TEMPEST" = "COMPASS_TEMPEST_viz.R",
  "COMPASS:Synoptic" = "COMPASS_Synoptic_metadata_and_viz.Rmd",
  "LTREB" = "LTREB_viz.R",
  "SMARTX" = "SMARTX_viz.R",
  "GENX" = "GENX_viz.R",
  "FATE" = "FATE_viz.R",
)


external_code <- knitr::purl(
  viz_script,
  output = tempfile(fileext = ".R"),
  documentation = 2
)
```

```
##   |                                                                      |
```

```
knitr::read_chunk(external_code)


###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_Cl.txt"
  raw_file_name_so4 = "Raw Data/COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_SO4.txt"

#file path and name of processed data file
  processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_SO4_MSM_GWI_202607.csv"

###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

#qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2026.csv"
```

##Set Up Code - constants and QAQC cutoffs

##Read in metadata

##Pulls in Project-Specific Metadata cleaning chunk

```
code <- knitr::knit_code$get("clean_metadata")
eval(parse(text = code))
```

1 Run Information

Run date: 2026-07-30
Run by: Zoe Read
Run notes:

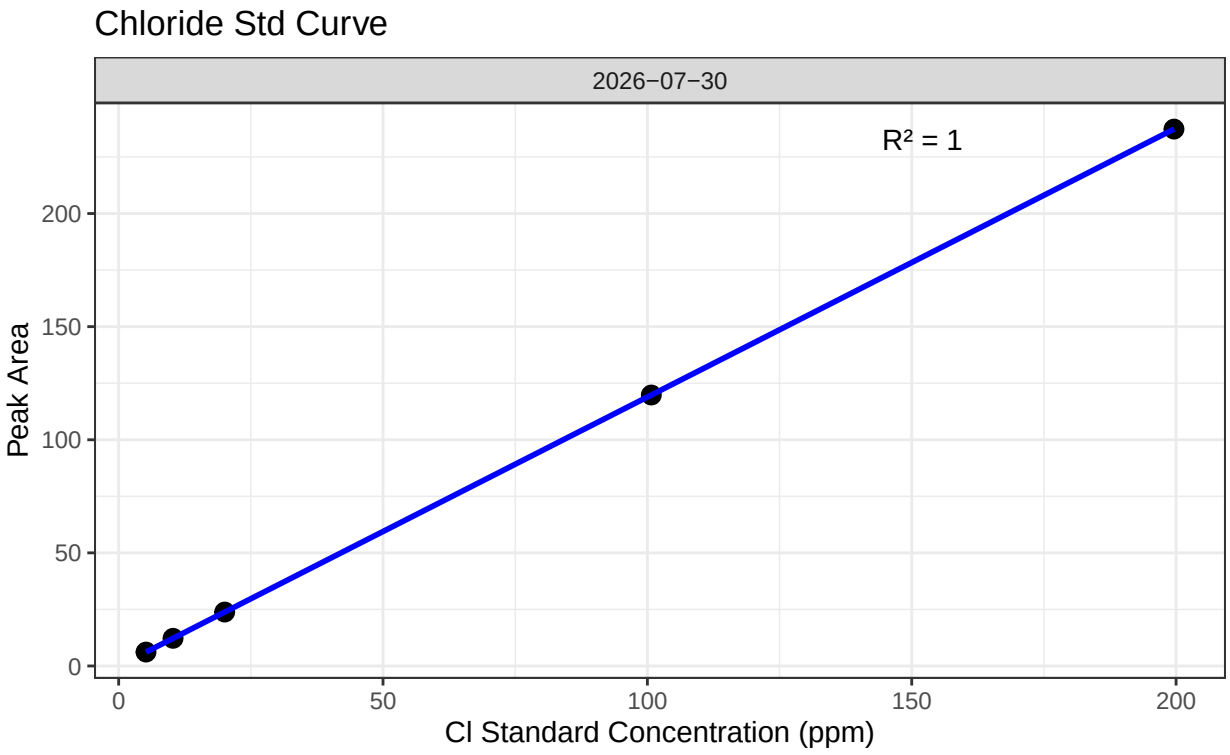
SO4 file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_SO4.txt

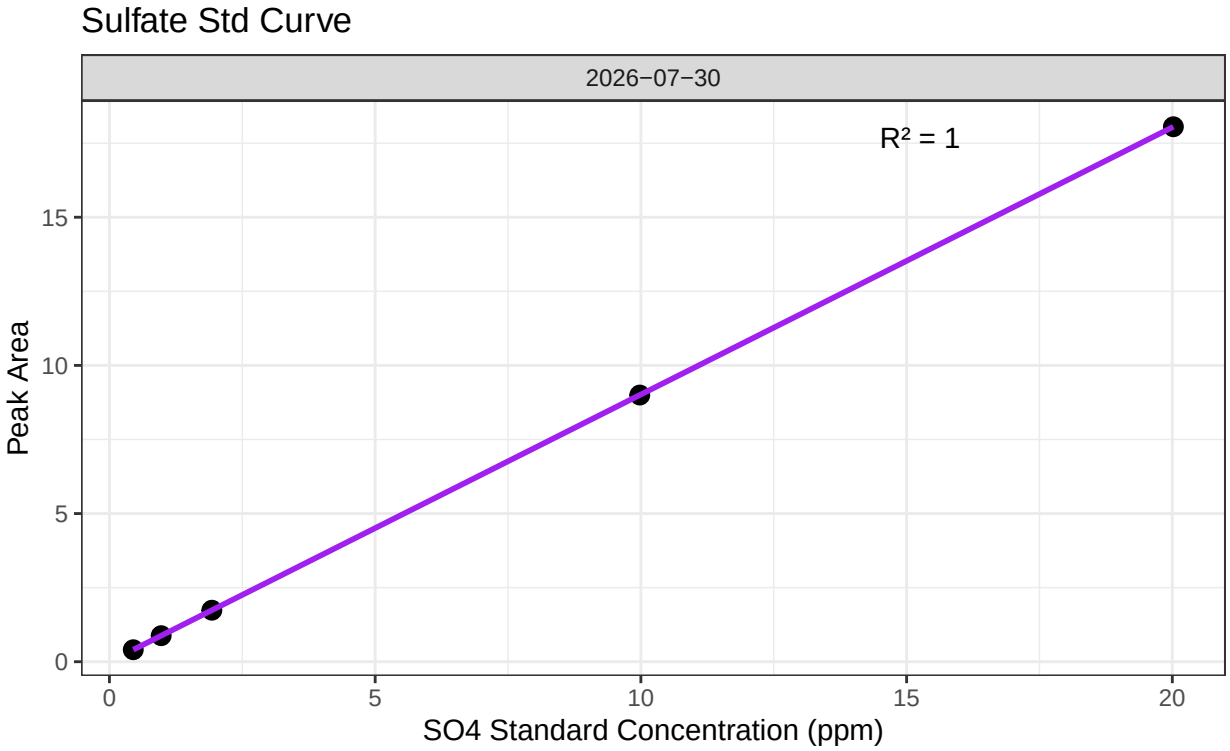
Cl file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_Cl.txt

##Import Sample Data

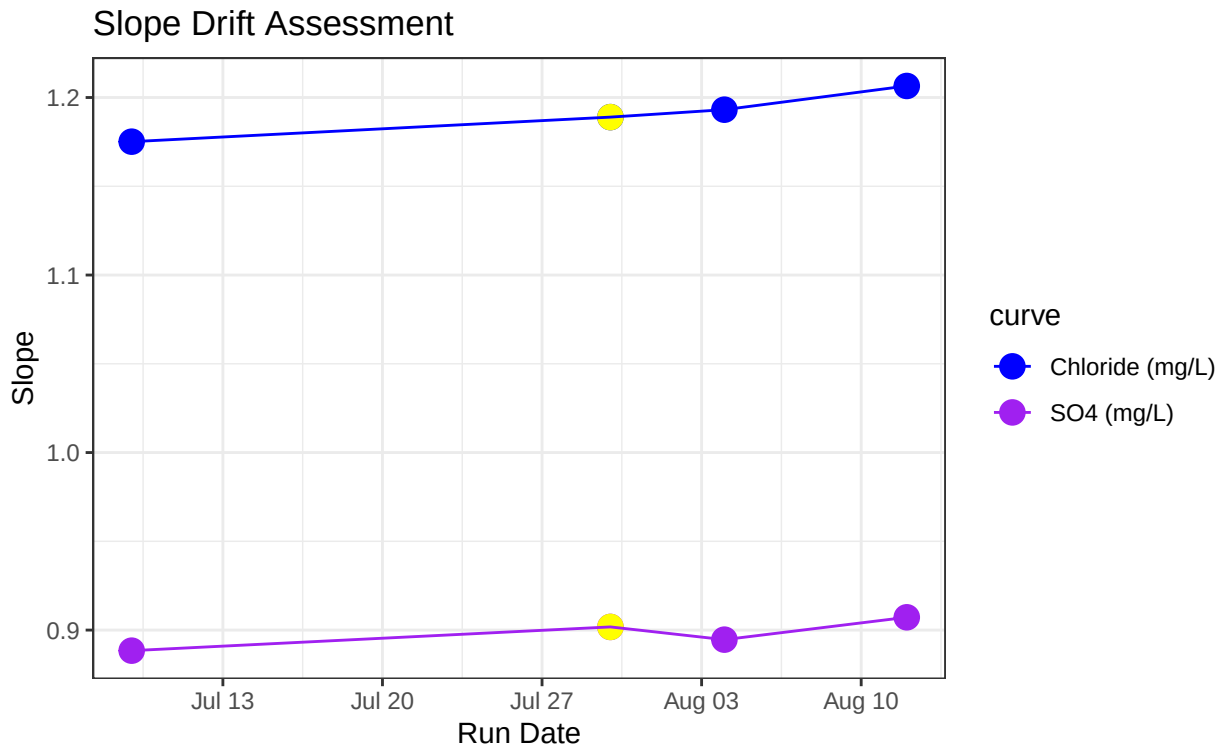
##Fix samples entered incorrectly

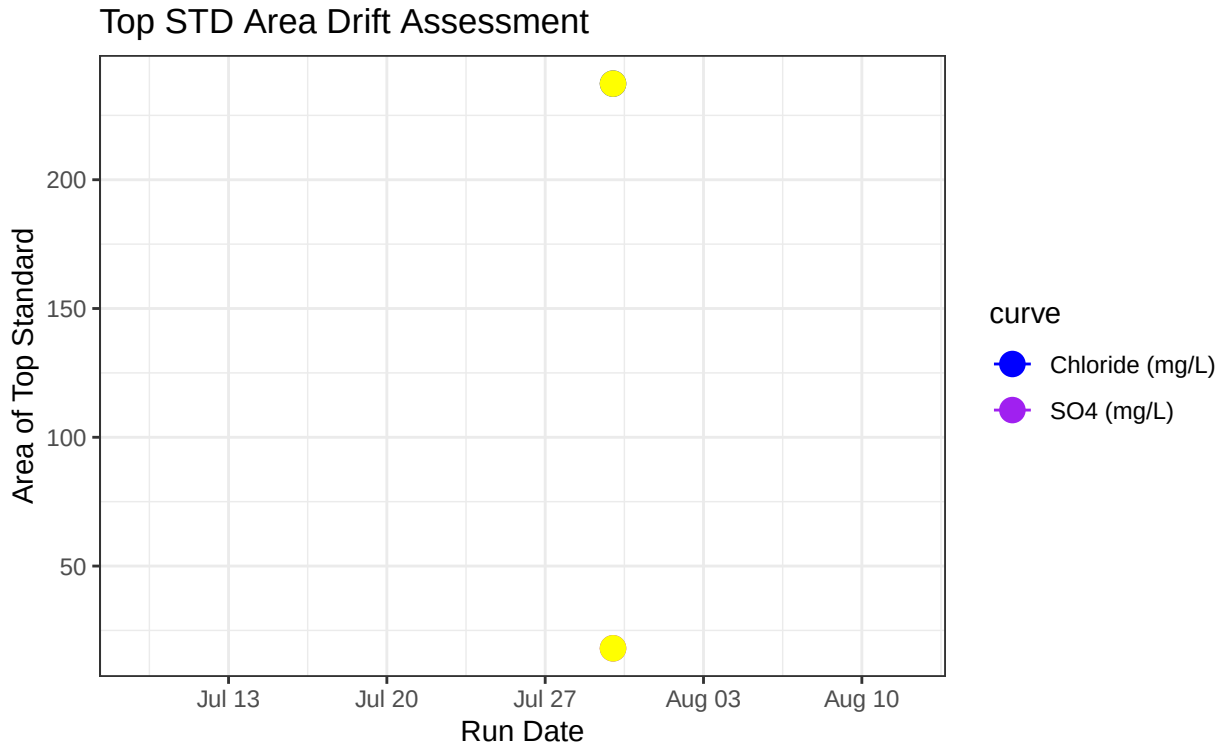
1.1 Assess Standard Curves






```
## [1] "QAQC log file exists and has been read into the code."
```






```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

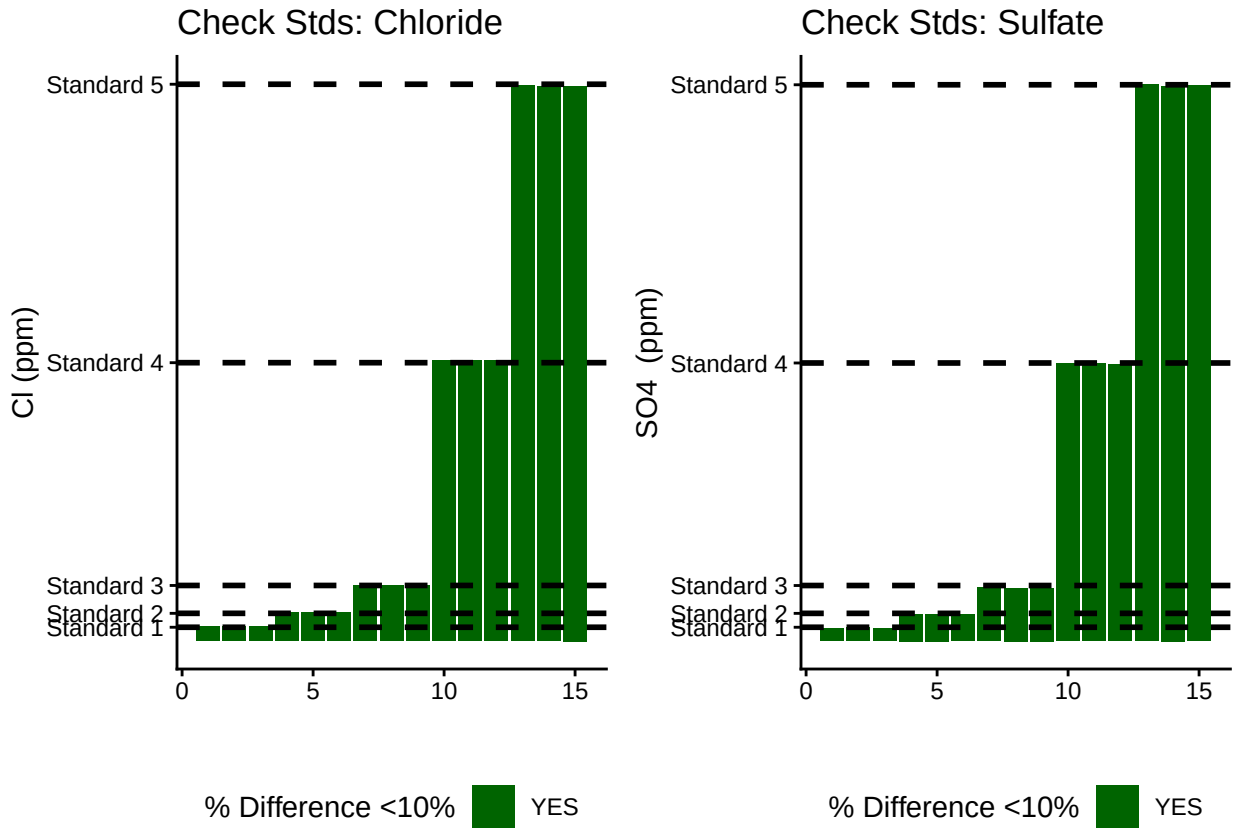
1.2 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl   sd_Cl    cv_Cl flag_Cl
##   <chr>        <dbl>   <dbl>    <dbl> <chr>
## 1 Standard 1    5.19 0.0218  0.00420  Chloride Check Standard RSD within Range ~
## 2 Standard 2   10.3  0.0192  0.00187  Chloride Check Standard RSD within Range ~
## 3 Standard 3   20.1  0.00968 0.000482 Chloride Check Standard RSD within Range ~
## 4 Standard 4  101.   0.0614  0.000609 Chloride Check Standard RSD within Range ~
## 5 Standard 5  199.   0.194   0.000972 Chloride Check Standard RSD within Range ~
```

```
## # A tibble: 5 x 5
##   sample_ID  mean_SO4  sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>   <dbl>   <dbl> <chr>
## 1 Standard 1    0.459 0.0195  0.0424  Sulfate Check Standard RSD within Range ~~
## 2 Standard 2    0.972 0.00658 0.00677 Sulfate Check Standard RSD within Range ~~
## 3 Standard 3    1.91  0.0114  0.00596 Sulfate Check Standard RSD within Range ~~
## 4 Standard 4    9.97  0.0165  0.00165 Sulfate Check Standard RSD within Range ~~
## 5 Standard 5   20.0   0.0312  0.00156 Sulfate Check Standard RSD within Range ~~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



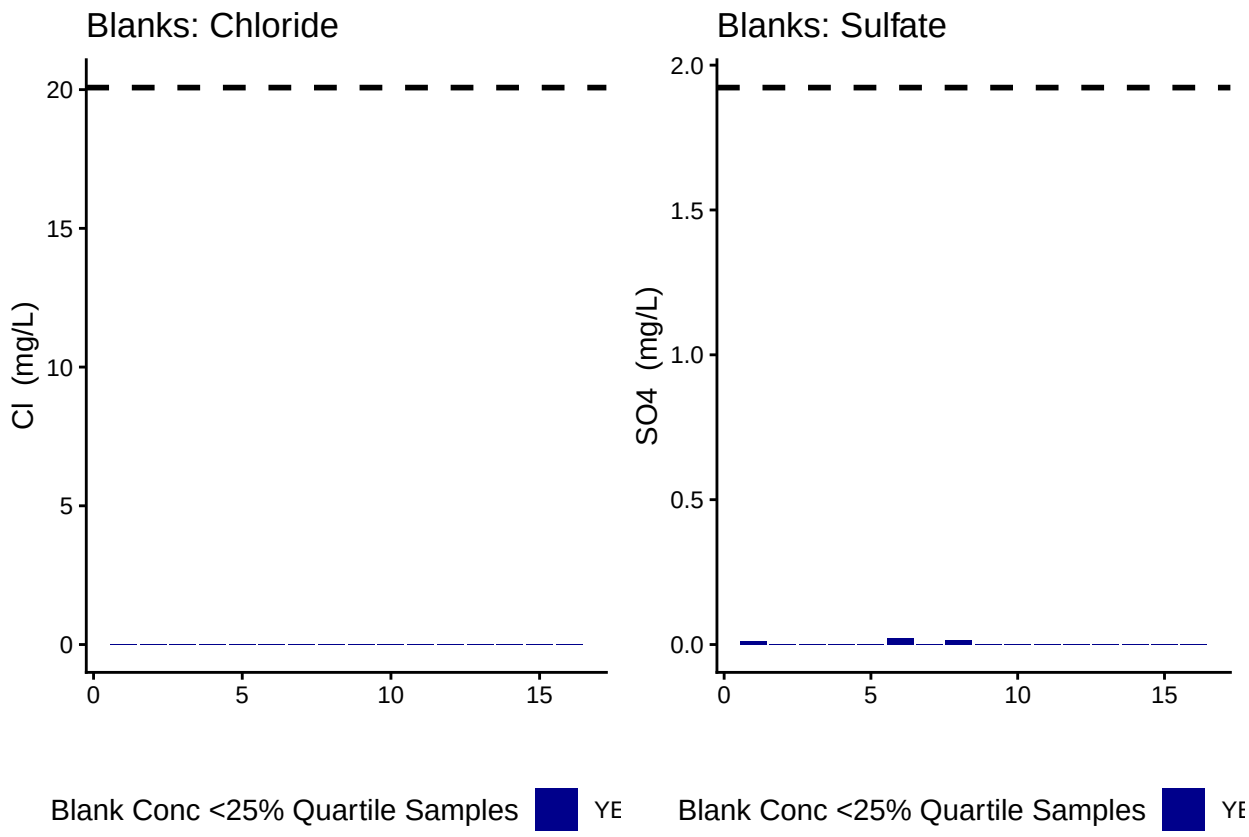

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

1.3 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



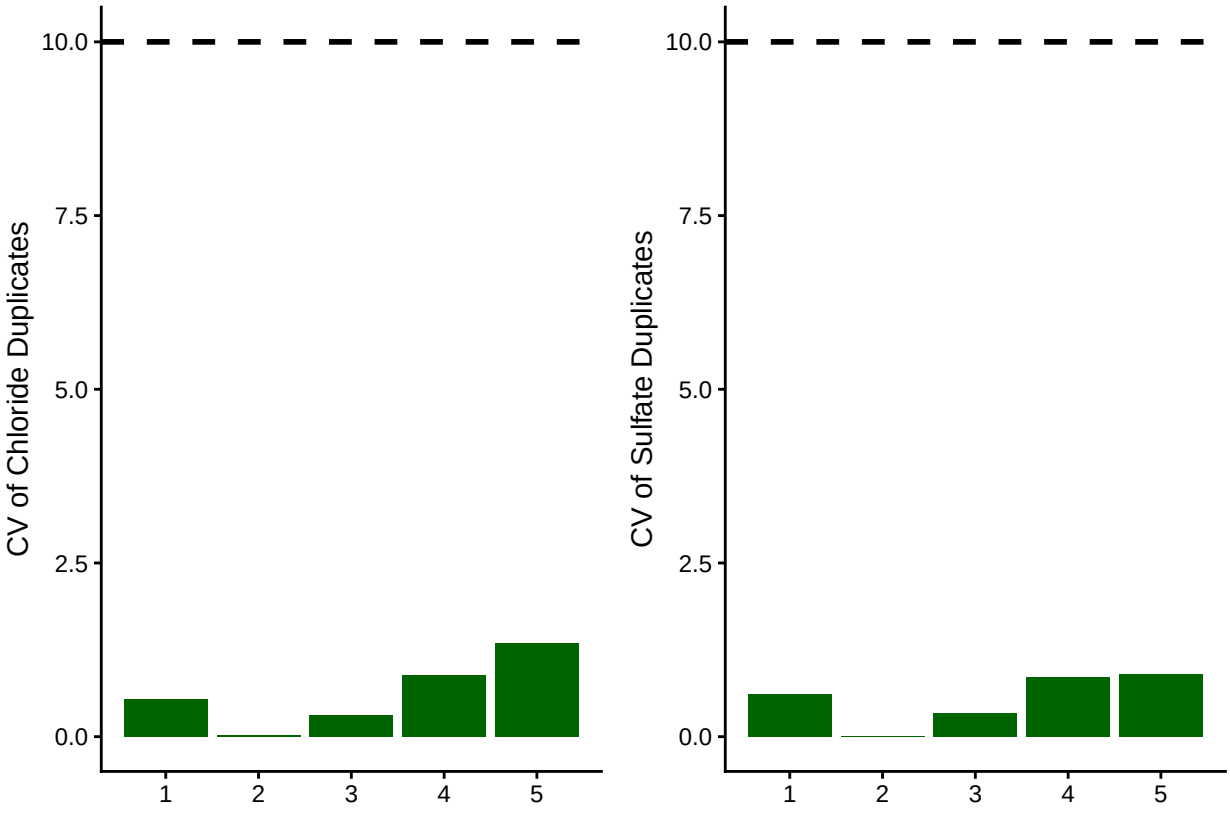

```
## Chloride blanks mean ppm:
```

```
## [1] 0.00820625
```

```
## Sulfate blanks mean ppm:
```

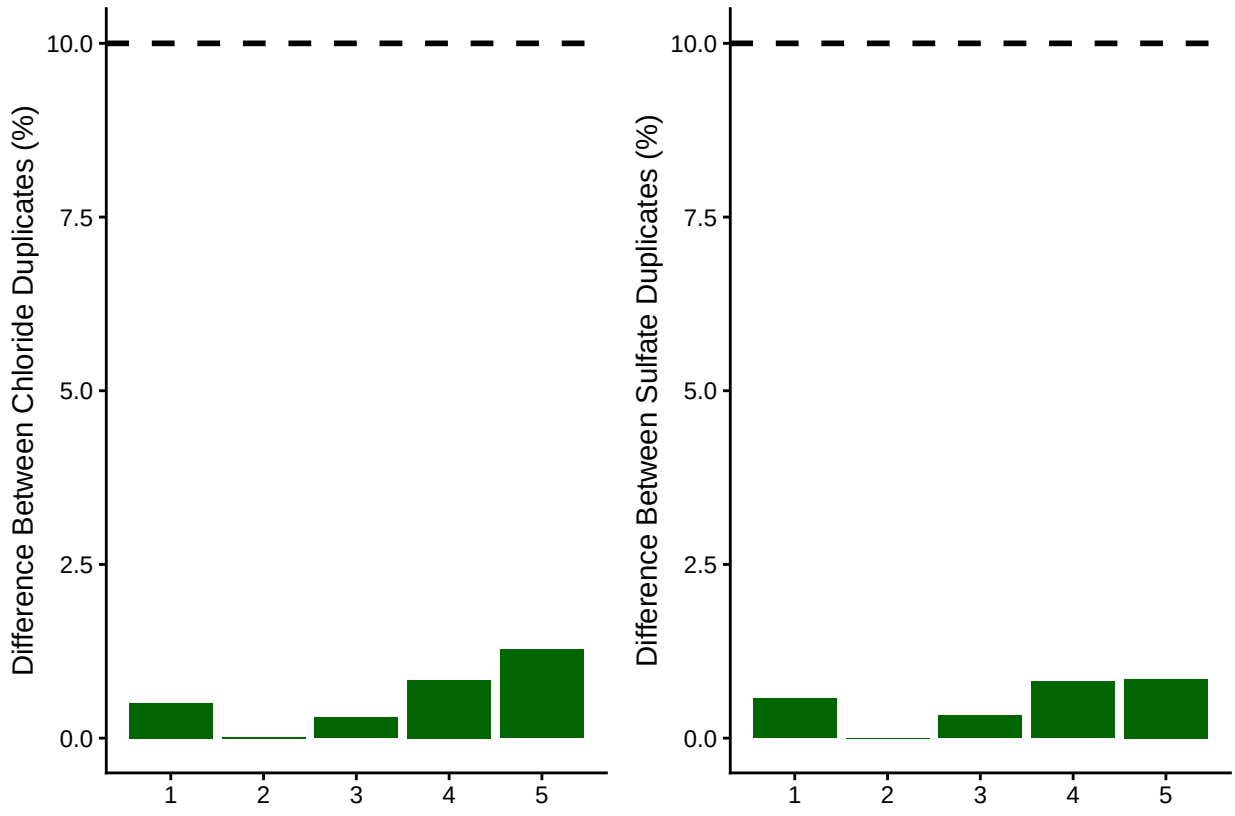
```
## [1] 0.003275
```

1.4 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```




```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

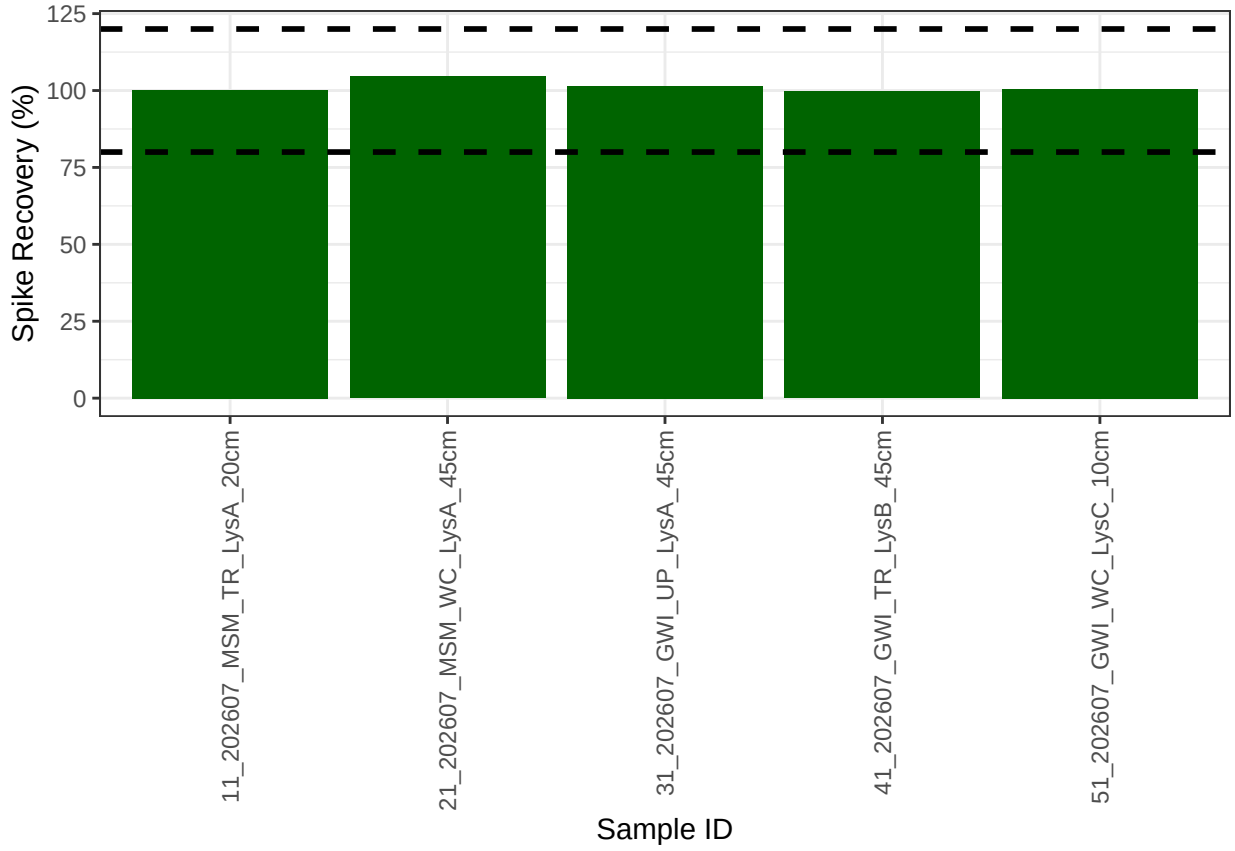
# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

##Pull in Dilution Lookup Table
dilutions <- read.csv("Dilution_Lookup_Table.csv")

dilutions <- dilutions %>%
  filter(Project == Project_name)
```

```r
##Add dilutions to dataframe
all_dat_dil <- all_dat %>%
  filter(sample_ID != "Lab Blank") %>%
  filter(!str_detect(sample_ID, "Standard")) %>%
  mutate(Site = ifelse(str_detect(sample_ID, Site_pattern), str_extract(sample_ID, Site_pattern), Site_
  mutate(Plot = ifelse(str_detect(sample_ID, Plot_pattern), str_extract(sample_ID, Plot_pattern), Plot_
  left_join(dilutions, by = c("Site", "Plot"))
```

1.6 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

1.7 Check if samples within the range of the standard curve

```
## Sample Flagging
```

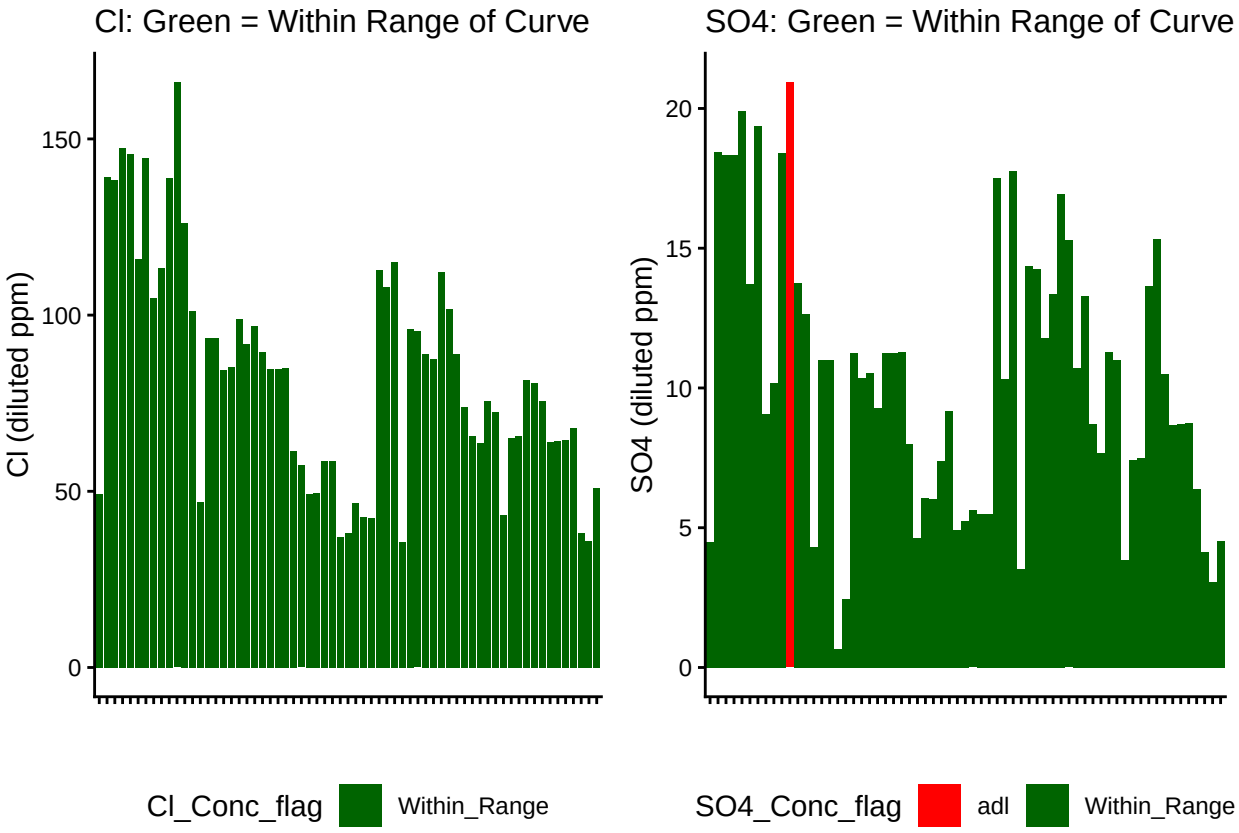


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	98.461539
adl	1.538461

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

1.8 Check to see if samples run match metadata & merge info

```
## Some sample IDs are missing from metadata.

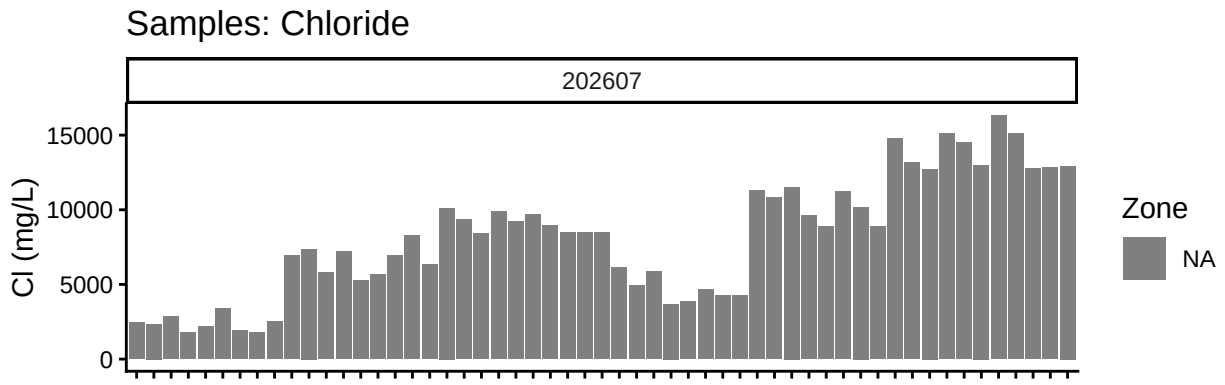
##  [1] "1_202607_MSM_UP_LYSA_10CM"  "2_202607_MSM_UP_LYSA_20CM"
##  [3] "3_202607_MSM_UP_LYSA_45CM"  "4_202607_MSM_UP_LYSB_10CM"
##  [5] "5_202607_MSM_UP_LYSB_20CM"  "6_202607_MSM_UP_LYSB_45CM"
##  [7] "7_202607_MSM_UP_LYSC_10CM"  "8_202607_MSM_UP_LYSC_20CM"
##  [9] "9_202607_MSM_UP_LYSC_45CM"  "10_202607_MSM_TR_LYSA_10CM"
## [11] "11_202607_MSM_TR_LYSA_20CM" "12_202607_MSM_TR_LYSA_45CM"
## [13] "13_202607_MSM_TR_LYSB_10CM" "14_202607_MSM_TR_LYSB_20CM"
```

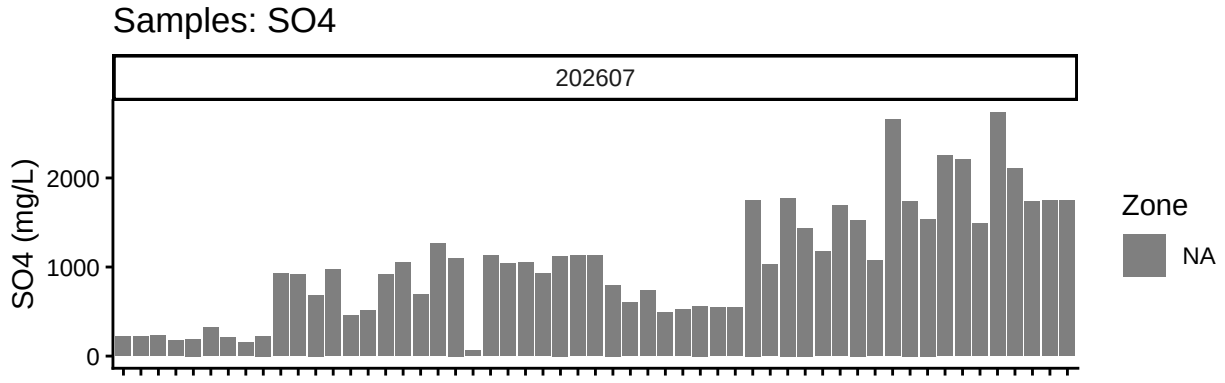
```
## [15] "15_202607_MSM_TR_LYSB_45CM" "16_202607_MSM_TR_LYSC_10CM"
## [17] "17_202607_MSM_TR_LYSC_20CM" "18_202607_MSM_TR_LYSC_45CM"
## [19] "19_202607_MSM_WC_LYSA_10CM" "20_202607_MSM_WC_LYSA_20CM"
## [21] "21_202607_MSM_WC_LYSA_45CM" "22_202607_MSM_WC_LYSB_10CM"
## [23] "23_202607_MSM_WC_LYSB_20CM" "24_202607_MSM_WC_LYSC_10CM"
## [25] "25_202607_MSM_WC_LYSC_20CM" "26_202607_MSM_SW_A"
## [27] "27_202607_MSM_SW_B"         "28_202607_MSM_SW_C"
## [29] "29_202607_GWI_UP_LYSA_10CM" "30_202607_GWI_UP_LYSA_20CM"
## [31] "31_202607_GWI_UP_LYSA_45CM" "32_202607_GWI_UP_LYSB_10CM"
## [33] "33_202607_GWI_UP_LYSB_20CM" "34_202607_GWI_UP_LYSB_45CM"
## [35] "35_202607_GWI_UP_LYSC_20CM" "36_202607_GWI_UP_LYSC_45CM"
## [37] "37_202607_GWI_TR_LYSA_10CM" "38_202607_GWI_TR_LYSA_20CM"
## [39] "39_202607_GWI_TR_LYSB_10CM" "40_202607_GWI_TR_LYSB_20CM"
## [41] "41_202607_GWI_TR_LYSB_45CM" "42_202607_GWI_TR_LYSC_10CM"
## [43] "43_202607_GWI_TR_LYSC_20CM" "44_202607_GWI_TR_LYSC_45CM"
## [45] "45_202607_GWI_WC_LYSA_10CM" "46_202607_GWI_WC_LYSA_20CM"
## [47] "47_202607_GWI_WC_LYSA_45CM" "48_202607_GWI_WC_LYSB_10CM"
## [49] "49_202607_GWI_WC_LYSB_20CM" "50_202607_GWI_WC_LYSB_45CM"
## [51] "51_202607_GWI_WC_LYSC_10CM" "52_202607_GWI_WC_LYSC_20CM"
## [53] "53_202607_GWI_SW_A"         "54_202607_GWI_SW_B"
## [55] "55_202607_GWI_SW_C"
```

1.9 Visualize Data by Plot - Pulls in viz_data code chunk from project-specific RMD

```
code <- knitr::knit_code$get("viz_data")
eval(parse(text = code))
```

```
## Warning: Expected 5 pieces. Additional pieces discarded in 49 rows [1, 2, 3, 4, 5, 6, 7,
## 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].
```





1.10 Clean Data - Pulls in clean_data code chunk from project-specific RMD

```
code <- knitr::knit_code$get("clean_data")
eval(parse(text = code))
```

1.11 Export Processed Data

#end